

Week 2

Identifying Data Requirements

1. Introduction

The first and most important step in designing a database is identifying the data requirements. Data requirements define what information the system needs to store, organize, retrieve, and manage efficiently.

For an E-Commerce Order Management System, the database must maintain information about customers, products, suppliers, orders, payments, shipments, and customer reviews. Proper identification of these data requirements ensures that the database is accurate, consistent, scalable, and capable of supporting business operations such as order processing, inventory management, payment tracking, and customer relationship management.

This document identifies all the major entities involved in the system, their attributes, and the primary keys required to uniquely identify each record.

2. Objectives

The objectives of identifying data requirements are:

- To determine all the information that must be stored in the database.
 - To identify the major entities involved in the e-commerce system.
 - To define the attributes associated with each entity.
 - To identify unique primary keys for every entity.
 - To identify foreign keys that establish relationships between entities.
 - To provide a strong foundation for designing the Entity Relationship (ER) Diagram and relational database schema.
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3. Data Requirements

The E-Commerce Order Management System requires the following major entities:

1. Customer
2. Product
3. Category
4. Supplier
5. Order
6. Payment
7. Shipment
8. Review

Each entity represents a real-world object and stores information necessary for the smooth operation of the online shopping platform.

4. Entity Details

4.1 Customer Entity

Description

The Customer entity stores all personal and contact information of users who register on the e-commerce website and purchase products.

Every customer has a unique identification number which distinguishes them from all other customers.

Attributes

Attribute	Description
Customer_ID (PK)	Unique identification number of the customer
First_Name	Customer's first name
Last_Name	Customer's last name
Gender	Gender of the customer
Date_of_Birth	Customer's date of birth
Email	Registered email address

Phone_Number	Contact number
Address	Residential address
City	City of residence
State	State of residence
Pincode	Postal code
Registration_Date	Date of account creation

Primary Key

Customer_ID

4.2 Product Entity

Description

The Product entity stores complete information about every product available for sale.

It helps customers browse products while enabling administrators to manage inventory and pricing.

Attributes

Attribute	Description
Product_ID (PK)	Unique product identification number
Product_Name	Name of the product
Description	Product details
Brand	Brand name
Price	Selling price
Stock_Quantity	Available inventory
Category_ID (FK)	Category to which product belongs
Supplier_ID (FK)	Supplier providing the product

Product_Image	Product image
Date_Added	Date product was added

Primary Key

Product_ID

4.3 Category Entity

Description

The Category entity groups similar products into different classifications such as Electronics, Clothing, Grocery, Furniture, etc.

Categories make product searching and browsing easier.

Attributes

Attribute	Description
Category_ID (PK)	Unique category identification number
Category_Name	Name of category
Description	Category description

Primary Key

Category_ID

4.4 Supplier Entity

Description

The Supplier entity stores information about companies or vendors supplying products to the e-commerce platform.

Attributes

Attribute	Description
Supplier_ID (PK)	Unique supplier identification number
Supplier_Name	Supplier company name
Contact_Person	Supplier representative
Phone_Number	Contact number
Email	Email address
Address	Office address
City	City
State	State

Primary Key

Supplier_ID

4.5 Order Entity

Description

The Order entity records every purchase made by customers.

Each order contains details about the customer, order date, order status, and total amount.

Attributes

Attribute	Description
Order_ID (PK)	Unique order identification number
Customer_ID (FK)	Customer placing the order
Order_Date	Date of purchase
Total_Amount	Total order value
Order_Status	Pending, Confirmed, Shipped, Delivered, Cancelled

Primary Key

Order_ID

4.6 Payment Entity

Description

The Payment entity stores information related to payments made for customer orders.

It enables tracking of successful, failed, and pending transactions.

Attributes

Attribute	Description
Payment_ID (PK)	Unique payment identification number
Order_ID (FK)	Related order
Payment_Method	UPI, Credit Card, Debit Card, Net Banking, Cash on Delivery
Payment_Date	Date of payment
Amount	Payment amount
Transaction_ID	Transaction reference number
Payment_Status	Success, Failed, Pending

Primary Key

Payment_ID

4.7 Shipment Entity

Description

The Shipment entity stores delivery information after an order has been dispatched.

Customers can use shipment details to track their orders.

Attributes

Attribute	Description
Shipment_ID (PK)	Unique shipment identification number
Order_ID (FK)	Related order
Courier_Name	Delivery service provider
Tracking_Number	Shipment tracking number
Shipment_Date	Date of dispatch
Expected_Delivery_Date	Estimated delivery date
Delivery_Date	Actual delivery date
Shipment_Status	Packed, Shipped, In Transit, Delivered

Primary Key

Shipment_ID

4.8 Review Entity

Description

The Review entity stores customer feedback after purchasing products.

Reviews improve customer trust and help other buyers make informed purchasing decisions.

Attributes

Attribute	Description
Review_ID (PK)	Unique review identification number
Customer_ID (FK)	Customer providing review
Product_ID (FK)	Reviewed product
Rating	Product rating (1–5)
Review_Text	Customer comments

Review_Date	Date review was submitted
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Primary Key

Review_ID

5. Primary Keys

A Primary Key uniquely identifies each record in a database table. No two records can have the same primary key value, and it cannot be NULL.

Entity	Primary Key
Customer	Customer_ID
Product	Product_ID
Category	Category_ID
Supplier	Supplier_ID
Order	Order_ID
Payment	Payment_ID
Shipment	Shipment_ID
Review	Review_ID

6. Foreign Keys

Foreign Keys establish relationships between entities and maintain referential integrity.

Entity	Foreign Key	References
Product	Category_ID	Category

Product	Supplier_ID	Supplier
Order	Customer_ID	Customer
Payment	Order_ID	Order
Shipment	Order_ID	Order
Review	Customer_ID	Customer
Review	Product_ID	Product

7. Entity Relationships

The relationships among the entities are as follows:

- One customer can place many orders.
 - One order belongs to one customer.
 - One order has one payment.
 - One order has one shipment.
 - One supplier can supply many products.
 - One category contains many products.
 - One customer can write multiple reviews.
 - One product can receive multiple reviews.
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8. Importance of Identifying Data Requirements

Identifying data requirements before designing a database provides several advantages:

- Ensures all business information is captured accurately.
- Eliminates unnecessary or duplicate data.
- Improves data consistency and integrity.
- Simplifies future database design and implementation.
- Supports efficient querying and reporting.
- Reduces redundancy through proper normalization.
- Helps create meaningful relationships among entities.
- Improves overall database performance and maintainability.

9. Conclusion

Identifying the data requirements is the foundation of designing an effective database for an E-Commerce Order Management System. In this phase, the major entities, their attributes, primary keys, and foreign key relationships have been identified. These requirements ensure that the database can efficiently manage customer information, products, orders, payments, shipments, suppliers, and reviews. The documented data requirements will serve as the basis for creating the Entity Relationship (ER) Diagram, normalization process, relational schema, and SQL database implementation in the following stages of the project.